DSA06 DATA HANDLING AND VISUALIZATION | Activity: Map Storytelling Challenge | Slot D

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Time: 20 minutes Total: 20 marks Tool: R (ggplot2 + maps). Your map story is different from your neighbour's; in the last 3 minutes, swap stories with a partner and say what you would change.

Story Brief MS-D16: Murder rate (per 100,000, 1976) [20 marks]

Brief: You are building a map story about Murder rate (per 100,000, 1976) across the United States for a district collector's review meeting. Your comparison variable is Life expectancy (years, 1969-71). Your focus region is the West region. Your map uses 5 classes and the "mako" colour scale. Statistics use all 50 states; the map shows the 48 contiguous states (Alaska and Hawaii are not in map_data).

Set-up: run this code first (do not edit it).

```
library(ggplot2); library(maps)
d <- data.frame(state = state.name, zone = state.region, state.x77)
names(d) <- c("state", "zone", "pop", "income", "illit", "life", "murder", "hsgrad", "frost", "area")
d$id <- tolower(d$state)
mapdat <- merge(map_data("state"), d, by.x = "region", by.y = "id")
mapdat <- mapdat[order(mapdat$group, mapdat$order), ]
V <- d$murder      # the variable you map
W <- d$life      # the comparison variable
```

Tasks:

1. The base map. Draw a choropleth of murder: ggplot(mapdat, aes(long, lat, group = group, fill = murder)) + geom_polygon(colour = "white") + coord_map("albers", lat0 = 29.5, lat1 = 45.5) + scale_fill_viridis_c(option = "mako") with a clear title and legend [1]. Report the highest state and its value [1], the lowest value and the state(s) that have it [1], and the median of all 50 states [1]. [4]
2. The regional story. Use tapply(V, d$zone, mean). Give the mean for each of the four regions [1.5] and name the highest and lowest region and the gap between them [1]. Then compute cor(V, W), describe its strength (weak below 0.3, moderate 0.3 to 0.7, strong above 0.7) and direction in one sentence, noting that it is not proof of cause [1.5]. [4]
3. Classes change the story. Make two versions of the map with 5 classes: equal interval, cut(V, 5), and quantile, cut(V, quantile(V, seq(0, 1, length.out = 6)), include.lowest = TRUE). For each, report the lower limit of the top class and the number of states in it [2.5]. Which method makes the highest values look rarer, and which would you choose for your audience? [1.5] [4]
4. Counts versus rates. Create the number of murders: count <- d$pop * 1000 * d$murder / 100000. Name the top 3 states by count and the top 3 states by murder [2]; say how many states appear in both lists [1]. In one sentence, explain why a map of counts would mislead and what to map instead [1]. [4]
5. Tell the story for the West region. Report the mean of murder in West and its ratio to the mean of all 50 states [1]; the highest state in West with its value [0.5]; and how many West states are above the national median [0.5]. Then write a headline of at most 12 words [1], and two evidence sentences using your numbers plus one recommendation for a district collector's review meeting [1]. [4]

Good map stories: *one clear message, honest classes, rates not raw counts, a readable legend, and a source note.*

Code of Conduct: I certify that this work is my own. Signature: _____

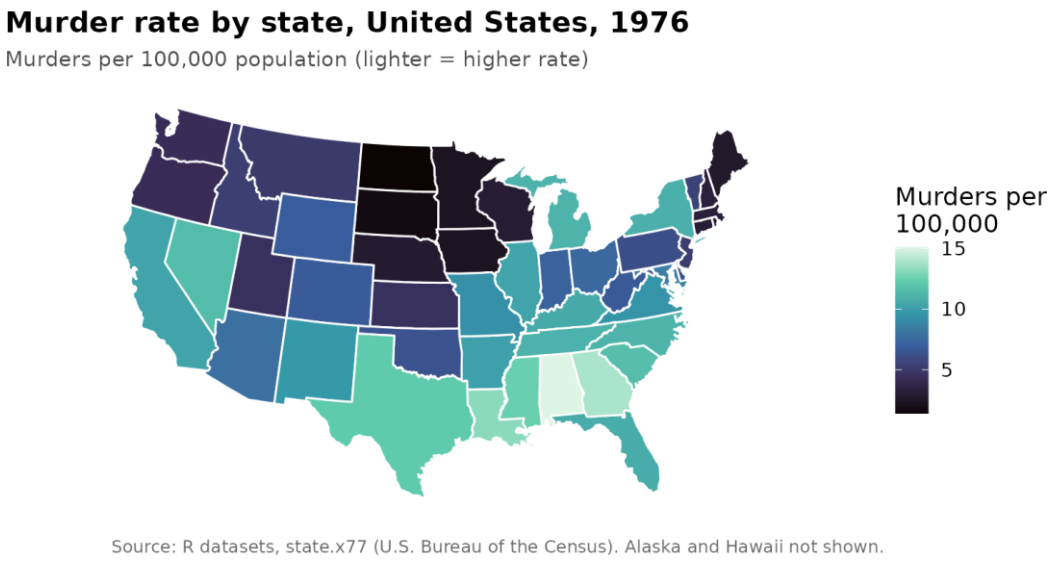
1.Question: *Draw a choropleth of murder with ggplot(mapdat, aes(long, lat, group = group, fill = murder)) + geom_polygon(colour = "white") + coord_map("albers", lat0 = 29.5, lat1 = 45.5) + scale_fill_viridis_c(option = "mako") with a clear title and legend. Report the highest state and its value, the lowest value and the state(s) that have it, and the median of all 50 states.*

R code

```
ggplot(mapdat, aes(x = long, y = lat, group = group, fill = murder)) +
  geom_polygon(colour = "white") +
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
  scale_fill_viridis_c(option = "mako") +
  labs(title = "Murder rate by state, United States, 1976",
       subtitle = "Murders per 100,000 population (lighter = higher rate)",
       fill = "Murders per\n100,000",
       caption = "Source: R datasets, state.x77 (U.S. Bureau of the Census)") +
  theme_void()

d$state[which.max(V)]; max(V)                    # highest
min(V); d$state[V == min(V)]                     # lowest
median(V)                                        # median of 50 states
```

Map



Quantity	Value
Highest state	Alabama, 15.1 murders per 100,000
Lowest value	1.4 per 100,000, in North Dakota (the only state with this value)
Median of all 50 states	6.85 per 100,000

2.Question: *Use tapply(V, d$zone, mean). Give the mean for each of the four regions and name the highest and lowest region and the gap between them. Then compute cor(V, W), describe its strength and direction in one sentence, noting that it is not proof of cause.*

R code

```
tm <- tapply(V, d$zone, mean)
round(tm, 2)
names(which.max(tm)); names(which.min(tm))
round(max(tm) - min(tm), 2)         # gap between highest and lowest region

round(cor(V, W), 2)                  # murder rate vs life expectancy
```

Regional means of the murder rate

Northeast	South	North Central	West
4.72	10.58	5.28	7.22

Quantity	Value
Highest region	South, mean 10.58 per 100,000
Lowest region	Northeast, mean 4.72 per 100,000
Gap (highest minus lowest)	10.58 - 4.72 = 5.86 per 100,000 (the South is about 2.2 times the Northeast)
cor(V, W)	-0.78

Strength and direction: r = -0.78 is a strong negative correlation (above 0.7 in size): states with higher murder rates tend to have lower life expectancy, but this is not proof of cause, because other factors such as poverty, health care and education could drive both.

3.Question: *Make two versions of the map with 5 classes: equal interval, cut(V, 5), and quantile, cut(V, quantile(V, seq(0, 1, length.out = 6)), include.lowest = TRUE). For each, report the lower limit of the top class and the number of states in it. Which method makes the highest values look rarer, and which would you choose for your audience?*

R code

```
d$eq <- cut(V, 5)                                                    # equal interval
d$qt <- cut(V, quantile(V, seq(0, 1, length.out = 6)), include.lowest = TRUE)  # quantile
table(d$eq); table(d$qt)

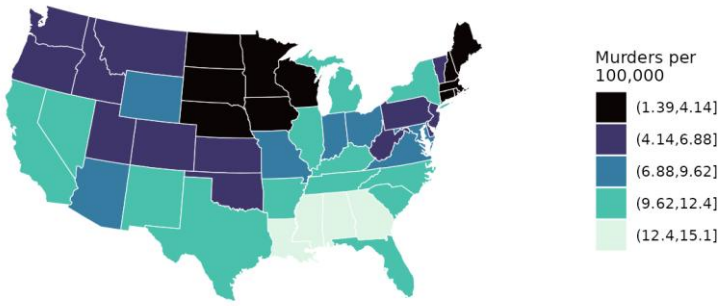
# lower limit of the top class and the number of states in it
1.4 + 4 * (max(V) - min(V)) / 5; sum(d$eq == levels(d$eq)[5])
quantile(V, 0.8);                sum(d$qt == levels(d$qt)[5])

md <- merge(map_data("state"), d, by.x = "region", by.y = "id")
md <- md[order(md$group, md$order), ]
ggplot(md, aes(long, lat, group = group, fill = eq)) +     # repeat with fill = qt
  geom_polygon(colour = "white", linewidth = 0.2) +
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
  scale_fill_viridis_d(option = "mako", name = "Murders per\n100,000") + theme_void()
```

The two maps

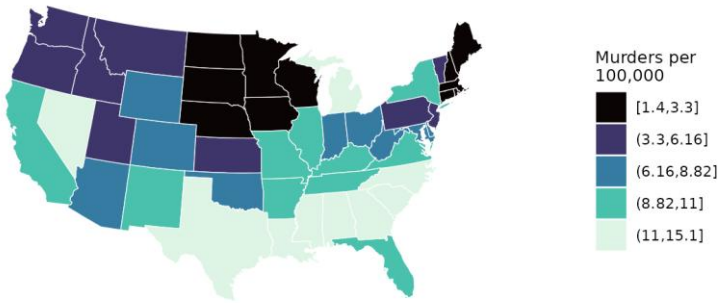
A. Equal interval: cut(V, 5)

5 classes of equal width



B. Quantile: cut(V, quantile(V, seq(0,1,length.out=6)))

5 classes with about 10 states each



Method	Class counts (low to high)	Lower limit of top class	States in top class	Share of states
Equal interval	11, 14, 7, 14, 4	12.36	4	8%
Quantile	11, 9, 10, 10, 10	11.02	10	20%

The equal-interval top class holds only Alabama (15.1), Georgia (13.9), Louisiana (13.2) and Mississippi (12.5). The quantile top class also adds Texas (12.2), South Carolina (11.6), Nevada (11.5), Alaska (11.3), Michigan (11.1) and North Carolina (11.1).

Which makes the highest values look rarer: Equal interval. Its top class covers only the extreme end of the range, so just 4 of 50 states look “high”, whereas the quantile map shades 10 states in the top colour.

My choice: Quantile. For a district collector who wants to see which states are in the worst fifth, equal counts per class spread the colours evenly and make the ranking easy to read; the class limits are shown in the legend, so the small differences near the class edges (for example 11.1 versus 15.1 in the top class) can still be seen.

4.Question: *Create the number of murders: count <- d$pop * 1000 * d$murder / 100000. Name the top 3 states by count and the top 3 states by rate; how many states appear in both lists? In one sentence, explain why a map of counts could mislead and say which map you would choose instead.*

R code

```
d$count <- d$pop * 1000 * d$murder / 100000
top_count <- d[order(-d$count)[1:3], c("state", "count", "murder")]
top_rate  <- d[order(-d$murder)[1:3], c("state", "count", "murder")]
top_count; top_rate
intersect(top_count$state, top_rate$state)       # states in both lists
length(intersect(top_count$state, top_rate$state))
```

Rank	Top 3 by count	Murders (count)	Top 3 by rate	Rate per 100,000
1	California	2,183	Alabama	15.1
2	New York	1,970	Georgia	13.9
3	Texas	1,493	Louisiana	13.2

States in both lists: 0. The two top-3 lists share no state.

Why counts mislead: A map of counts mostly shows population size, since California, New York and Texas have the most people, so it paints big states as the most dangerous and hides high-rate states such as Alabama and Louisiana; I would map the rate per 100,000, which compares states fairly.

Counts versus rates

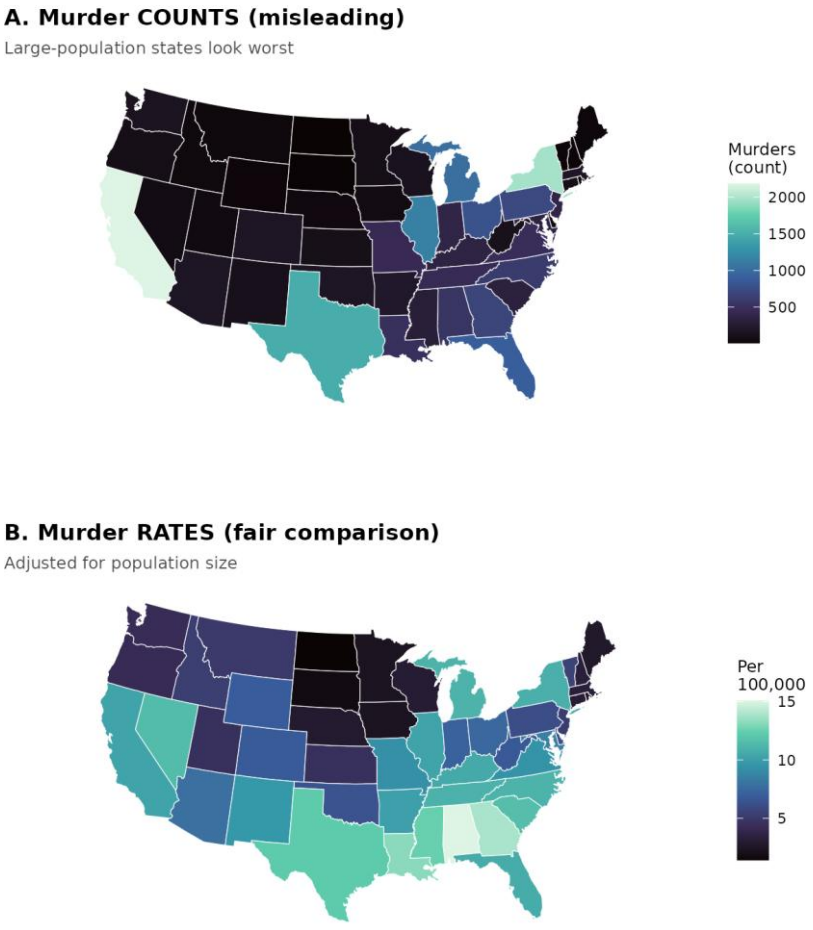


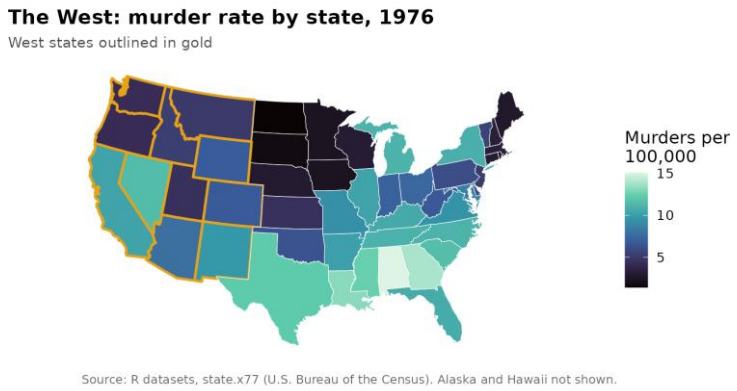
Figure 3: Map of counts (top) highlights California and New York; map of rates (bottom) highlights the Deep South.

5.Question: *Report the mean of murder in West and its ratio to the mean of all 50 states; the highest state in West with its value; and how many West states are above the national median. Then write a headline of at most 12 words, and two evidence sentences using your numbers plus one recommendation for a district collector's review meeting.*

R code

```
w <- d[d$zone == "West", ]            # 13 states
round(mean(w$murder), 2)                # West mean
round(mean(V), 2)                       # national mean
round(mean(w$murder) / mean(V), 2)      # ratio
w[which.max(w$murder), c("state", "murder")]
sum(w$murder > median(V))              # West states above the national median
w$state[w$murder > median(V)]
```

Map (West outlined in gold)



Quantity	Value
Mean murder rate in the West (13 states)	7.22 per 100,000
Mean of all 50 states	7.38 per 100,000
Ratio (West / all 50)	7.22 / 7.38 = 0.98
Highest state in the West	Nevada, 11.5 per 100,000
West states above the national median (6.85)	6 of 13: Nevada, Alaska, California, New Mexico, Arizona, Wyoming

Map story

Headline (11 words): West Mirrors National Murder Rate, but Nevada and Alaska Stand Out

Evidence 1: The West's mean murder rate is 7.2 per 100,000, almost the same as the national mean of 7.4 (a ratio of 0.98), and only 6 of its 13 states lie above the national median of 6.85.

Evidence 2: The average hides a wide spread, from 4.2 in Oregon to 11.5 in Nevada, and Nevada and Alaska (11.3) rank among the ten highest-rate states in the country, about 2.7 times the rate in Oregon.

Recommendation: Review the West state by state rather than by its regional average, and focus prevention effort on the states with rates above 10 per 100,000: Nevada, Alaska and California.